ASSEMBLY LANGUAGE PROGRAMS (USING MARIE)

OBJECTIVES

- The purpose of this applied session is getting to know MARIE assembly code programs.

INSTRUCTIONS

- Use MARIE simulator to write assembly language programs.
- You may work in a small group.

Assignment 1 Part 3 Quiz

Attempt the in-person Assignment 1 Part 3 Quiz in Moodle (only one attempt allowed).

Activity 1: MARIE Basics

Use the MARIE simulator to try out the program given below and answer the following questions. Copy the codes in the simulator input window, assemble your program to check for any errors, and then run or step through the lines of your codes to observe the changes taking place in CPU registers and various memory locations.

Line No.	Code	Comments
1	Load 004	/ Load value from address 004 into AC
2	Add 004	/ Add value from address 004 to value in AC
3	Store 004	/ Store AC into address 004
4	Halt	/ Stop execution
5	DEC 42	/ This is address 004, initialise with value 42

(a) Can you locate your program In MARIE memory? What are the memory addresses of the first and the last line of your program?
(b) In what format is your program stored in MARIE memory? How is your data “DEC 42” stored?
(c) What is the first and the last address of MARIE memory? Can you determine the size of MARIE memory?

(d) Using a variable to refer to “DEC 42”, we can modify the line no. 5 as follows:

5	num, DEC 42	/num is a variable and num = 42

Rewrite the program using the variable “num” instead of using the memory address to refer to the data “DEC 42.”

Activity 2: Data input, output and storage in MARIE

Write a MARIE program to input any data from the keyboard and display in the MARIE simulator output window. Assemble your program to check for any errors, and then run or step through the lines of your codes and answer the following questions.

(a) How many different input and output modes of data are available in MARIE?
(b) In which register of the CPU is your input data stored?
(c) Add the following instruction “ORG 020” at the beginning of your program, and assemble your code again. Now, you will find that MARIE has relocated your machine code to a new address. What is this address? Is it the same as the contents of the CPU register PC? Why? Step through your program to see the changes happening in the PC.
(d) Add two variables to your program as follows:

```
numIn,      DEC 0
One,        DEC 1
```

Note: add these lines at the end of your program.

Modify your program to store the input data in “numIn” and add ‘1’ to it before displaying it in the output. Please input decimal numbers (from the keyboard) for this task.

Activity 3: Branch Instructions in MARIE and use of labels

Extend your program from Activity 2, by adding labels, such as:

```
begin,      Load  num1
            Add   One
            ……….
```

(a) Further modify your program by adding jump instructions to repeat the data input, modify and display activities endlessly.
(b) Using “skipcond” instruction, exit the program when the user inputs data ‘0’.

Activity 4: Branch Instructions in MARIE and use of labels

A conditional statement allows the flow of a program to depend on data. In high-level programming languages, these are typically achieved using (i) if-then-else or (ii) while-do statements such as the following (in pseudocode):
(a) if-then-else

```
if (X > Y)
        X = X + X
else
        Y = Y + Y
```

(b) while-do

```
X = 1
while (X < 10) do
        X = X + 1
endwhile
```

Write MARIE assembly language programs implementing the conditional statements.

Post-class Activity: Reflection

Complete your weekly reflection activity in the Ed Lesson using the updated template (for your assignment 2). You MUST follow this template to write your reflections (**more than 200 words in each week**) to meet assignment hurdle requirement.